

Algebra PhD Exam, July 1997

Answer seven out of the following ten problems. State clearly any results you need.

1. Let R be a commutative ring with identity, and S a multiplicative system in R . Show that the localization $S^{-1}R$ is a flat R -module.
2. Show that the field of complex numbers has uncountably many distinct automorphisms.
3. This problem has two parts.
 - (a) Show that A_5 is simple.
 - (b) Use (a) to show that S_n is not solvable for $n \geq 5$.
4. If k is a field, show that the ring of power series $k[[T]]$ is a discrete valuation ring.
5. Suppose that p and q are distinct primes such that $q > p^2$ and $q \not\equiv 1 \pmod{p}$. Show that if G is a group of order p^2q , then G is abelian.
6. Let R be a local Noetherian (commutative) ring, and let M be a finitely generated R -module. Show that M is flat if and only if it is free.
7. Suppose that R is a ring with the property that every short exact sequence of unital R -modules splits. Show that every unital R -module is a direct sum of simple unital R -modules.
8. This problem has two parts.
 - (a) Show that if k is a perfect field, then an irreducible polynomial in $k[T]$ has no multiple roots. (Hint: show that if $f(T)$ is irreducible and has multiple roots, then the characteristic of k is $p > 0$, and $f(T) = g(T^p)$ for some $g \in k[T]$.)
 - (b) Give an example of a (non-perfect) field k and an irreducible polynomial $f(T)$ with multiple roots.
9. Suppose R is a ring with identity. For each left R -module M , show that $\text{Hom}_R(R, M)$ is naturally isomorphic to M (in the process, explain what “naturally isomorphic” means).
10. Classify up to isomorphism all semisimple rings of order 1584.